WEEK 12

Going Live and Wrapping Up

The system is online, the assistant lists records and class rank correctly, the code is cleaned up and the project is ready for GitHub

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This week we worked on four lines

1 Going live

Docker Compose on a Contabo VPS; HTTPS through Caddy; separate Postgres; .env only on the server.

smartschoolassistant.com is up

2 Assistant

get_risk tool, strict number rules, list mode and class rank for teachers (class_rank).

It can no longer drop a student

3 Interface

Grades table, weekly reports page, risk reasons in both languages, mobile selectors.

No keyboard on phones

4 Cleanup and GitHub

Silent error swallowing removed, AI tool files deleted, README written in TR and EN.

Clean repo: 0 secrets

The system is online: one door, four services

How we set it up

- One image (python:3.14.9-slim, installed exactly from the lock file); backend and frontend share it.
- Only Caddy is exposed (80/443); backend, frontend and database stay on the internal network.
- Same versions as the local test setup: Python 3.14.9, PostgreSQL 18. Cursor had written postgres:16; fixed.
- .env never enters the image; passwords and SECRET_KEY were regenerated on the server.

1,440
grade records (server)

6,444
attendance rows

5
RLS policies

80 · 443
the only open door

```
smartschoolassistant.com · Contabo VPS
postgres:18.6 · caddy:2.10.0
DEMO_TODAY 2026-11-09
```

Update flow and the md5 lesson

Step	What happens
1. Local tests	Full pytest run; the count comes from the raw output
2. rsync	Excluding .env, .venv, raw data and presentations
3. md5 diff	Per file; sorted with LC_ALL=C; the output must say AYNI
4. build + up	Backend and frontend again; never init (it wipes data)
5. enable_rls	Only if a database function changed; from inside backend

Two lessons

An && chain stopped silently when one step failed and the server kept old files; now every command ends with a confirmation word.

The md5 total differed between two machines although the files were the same: sort followed the locale. Sorting is now LC_ALL=C and the comparison is per file.

Assistant: a tool, a rule and a new function

get_risk

Calls the risk page's code with the same session. A class teacher only sees their own class; parents and students do not have the tool.

No copied calculation

Number rules

Numbers exactly as in the result; no ranges, rounding or "about". When the model wrote "between 71 and 74", the check caught it.

The check was not loosened

class_rank

"What is my class's rank in our school?" is answered now: rank, number of classes, two averages. No other class's name is returned.

The answer notes grade levels

class_rank is student_rank's counterpart for classes: it reads the class from the session, is SECURITY DEFINER and returns a row only for the teacher role.

List mode: the model cannot drop a student it never sees

Before

- The query returned 60 rows; the model received the first 50.
- The answer said "your 6 students" but listed 5; Zeynep Demir was missing.
- The grounding check could not catch it: every number written was in the result.

After

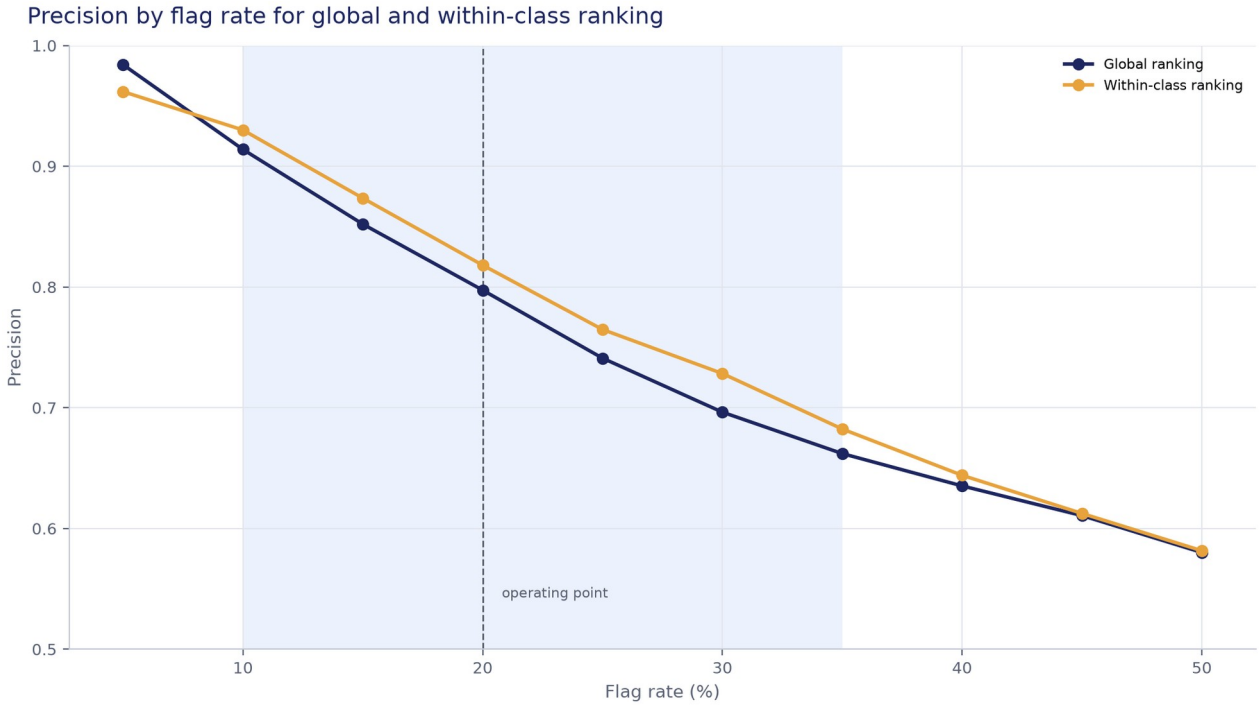
- run_sql got a presentation="table" option.
- In this mode the model sees only a summary: row_count, distinct_counts, date_range.
- The app draws the full table: one row per student, one column per exam, the date in the header.

A class of 32 gets the same two-sentence answer as a class of 6, and the table is complete.

What changed in the interface

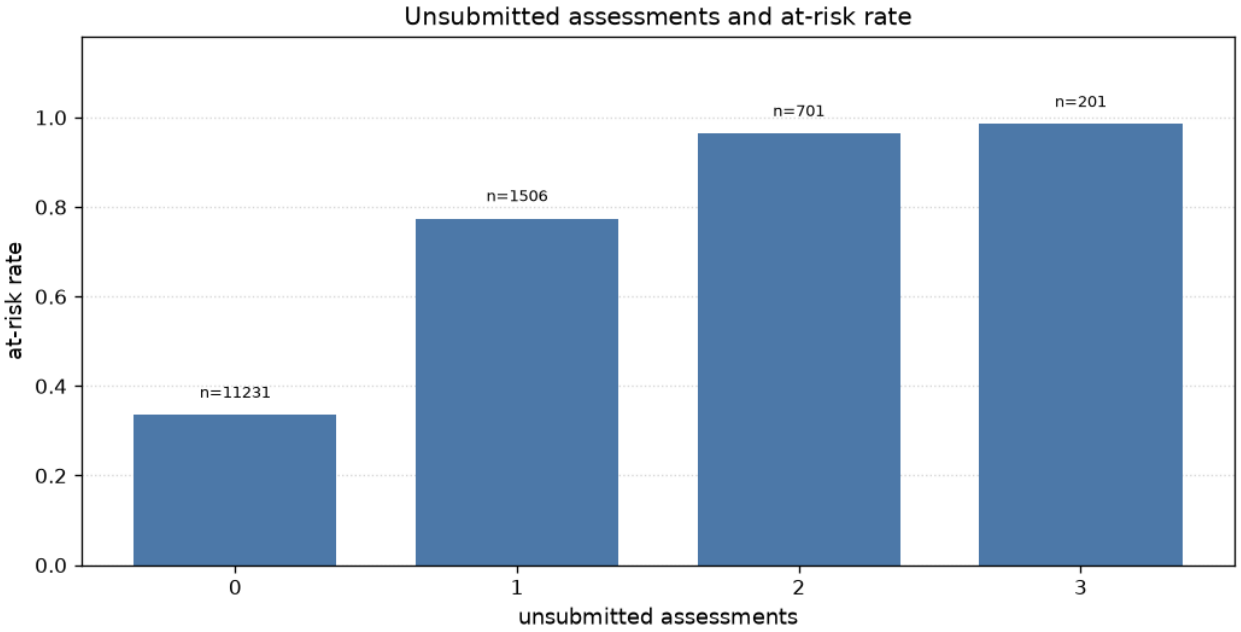
Topic	Before	After
Grades table	Names in a hidden index; one sparse row per date	One row per student, one column per exam; "Fizik 29.09"
Risk reasons	Turkish on the EN page too; "This is not a decision"	In the selected language; (tie) label
Weekly reports	Presentations only on my machine	Public page; slides and PDF
Mobile	Selectors opened the keyboard; top bar overflowed	Tap-only selection; demo chip hidden on phones
Research summary	Cards of different heights; reading time	Equal grid; data scale card

The research page reached its final form



- Figure: at 20%, within-class ranking is slightly more precise than global ranking; the school system's flag design rests on this.
- Summary cards sit in one grid with equal heights; the third card now shows the data scale.
- The reading time label and the "unknown" card were removed.
- Every number on the page still comes from the package; no hand-typed figures.

The signal the school data should add



- The model's strongest signal is the number of unsubmitted assignments.
- The school demo has no such record; the attention flag looks only at grades and absences.
- First item of future work: a homework submission record.

Cleanup and GitHub preparation

What we did

- main.py logs a warning instead of except: pass; explain.py has a single SHAP path; eval writes failed queries to stderr.
- .cursor, *.mdc and prompt files removed; tool wording cleaned from metin_yamasi.json.
- No raw data in the repo (studentVle.csv is 453 MB); the README gives the download steps.
- Weekly reports are in docs/haftalar; a test that read a file outside the repo now reads it from there.

63 MB
repository size

0
secret traces

21
weekly reports

2
READMEs (TR · EN)

Bug records (1/2)

Bug	Root cause	Fix
Tests and app in different environments	The environment was not documented; two .venvs	One environment, requirements.lock.txt
Mock gate does not protect the app	The gate uses cumulative risk, the app the term window	Temporary: fixed DEMO_TODAY; permanent fix open
_sor name clash	Cursor added a helper with the same name; 3 tests failed	New rule: no helper with an existing name
Blind asserts in weekly page tests	Proto name "imgs" always counted 0	at.get("image"); root from an env variable
Roles and GRANTs not in code	They were set up by hand in the local database	bootstrap_roles.py + sema.sql equality test
Assistant wrote a range	A rounded range; "71" was ungrounded	Instruction; check not loosened

Bug records (2/2)

Bug	Root cause	Fix
Server kept old files	The && chain stopped silently	Confirmation word + md5 diff
No names in the grades table	The pivot put names in the index, which was hidden	Name as a column; one column per exam
A student missing from a list	The model saw 50 of 60 rows	List mode: summary + full table
Risk reasons in Turkish on EN	The reason was built in one language in the backend	lang required; error on an unknown language
10 tests failed after a change	Two assess_risk callers were missed in the search	Search output read in full
A test fails on a clean clone	It read a file from ../week10	Source moved into docs/haftalar

Known limit: UCI notebook figures are byte-identical in TR and EN (same md5); the TR labels are in English.

Verification

1,366 passed

last full run (before the mobile fix)

```
grades|attendance|pg_policies
1440|6444|5

git ls-files: forbidden files 0
git grep sk-ant-: 0
total: 63M
```

- At the start of the week the full run was 1,335 passed; list mode, class_rank and risk language tests were added during the week.
- On the server, init set up the role, RLS and policies; four services are up, backend and db healthy.
- The package is unchanged: arastirma.json md5 27f5279b8b574814c2682d070603798e.
- Every deploy was checked with a per-file md5 diff.

Still open, and what comes next

Still open

- Rotate the API key; spending limit
- Login counter in process memory; question length limit, log masking
- Test the mock gate against the term window
- Regenerate sema.sql; image size (jupyter/shap)

Future work

- Homework submission record: the research's strongest signal
- A comparison-free flag for parents
- Subgroup audit
- The data needed to study teacher effects